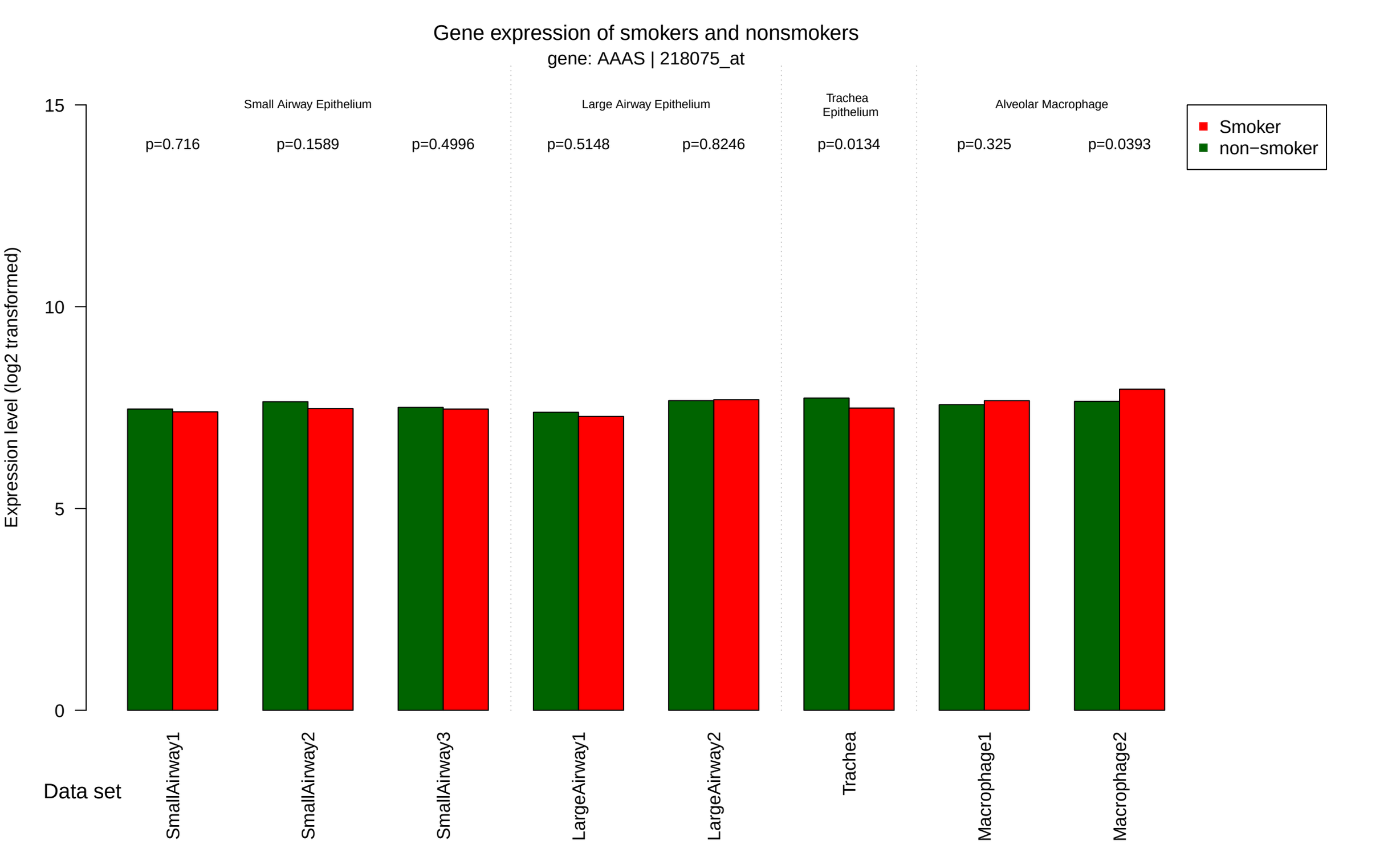
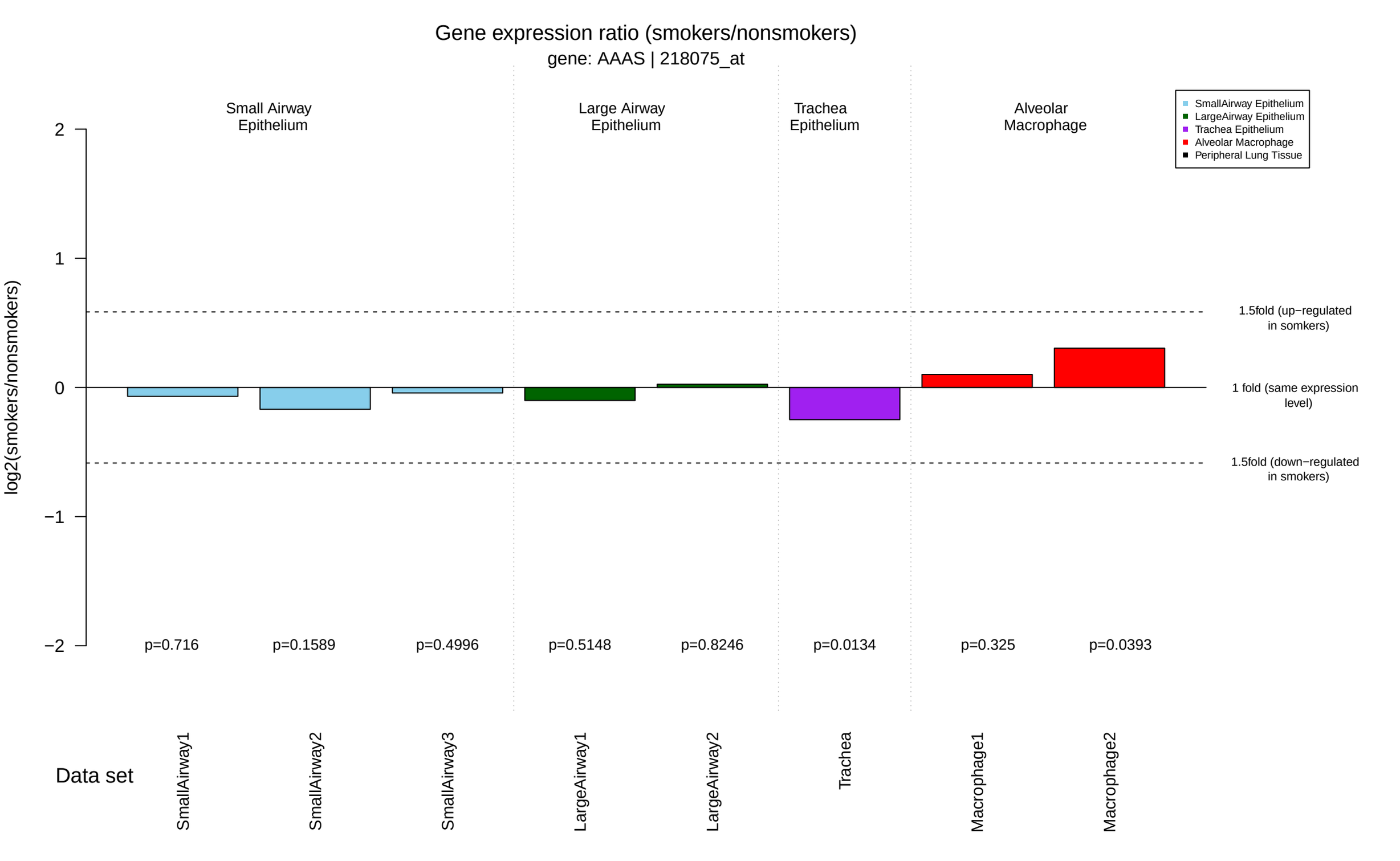
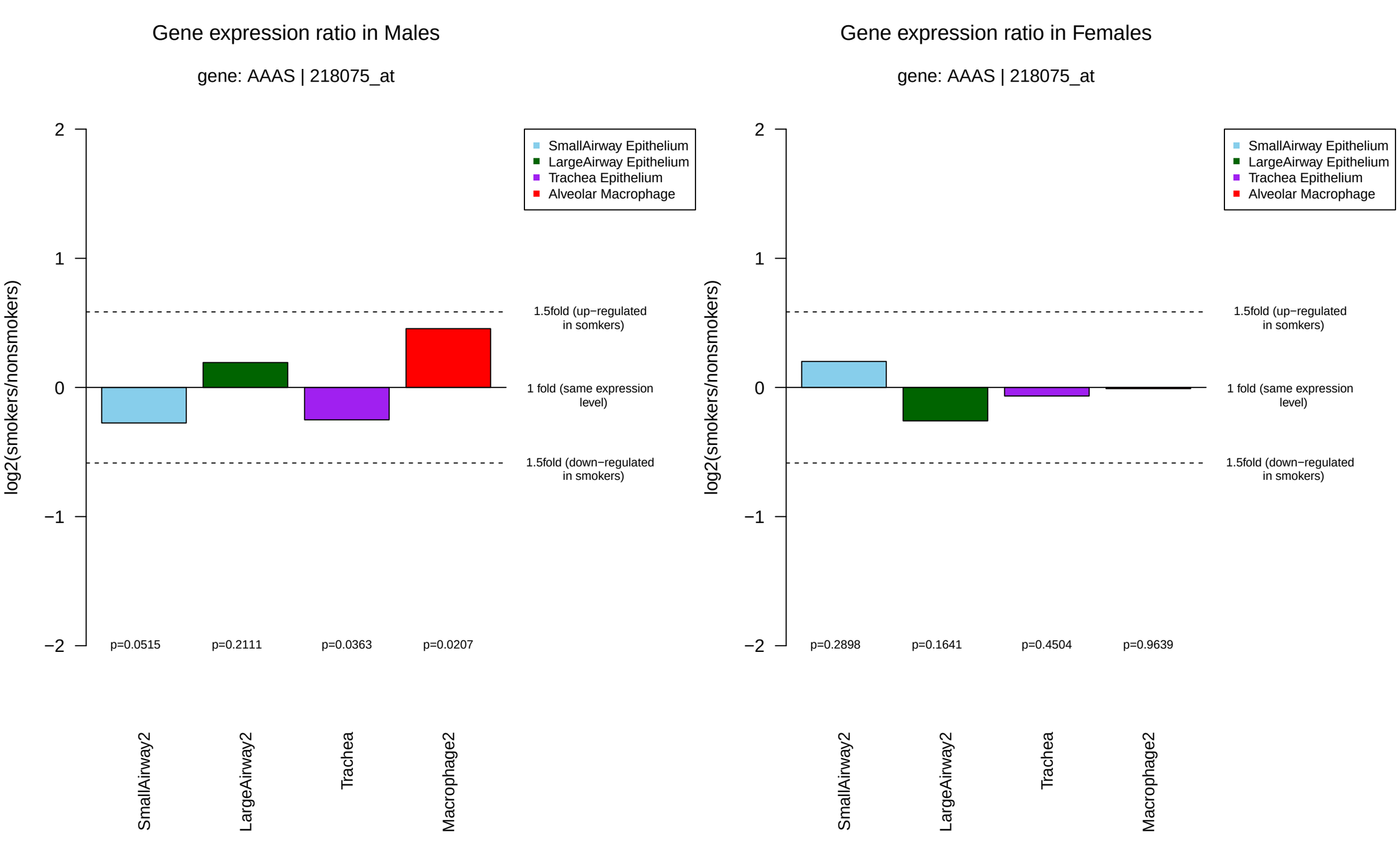
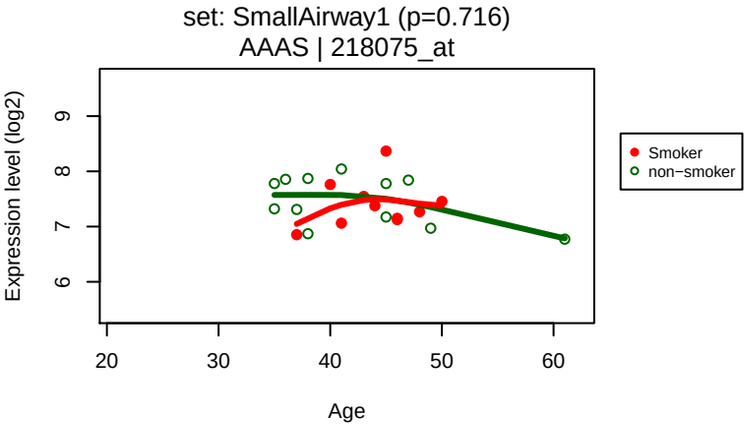
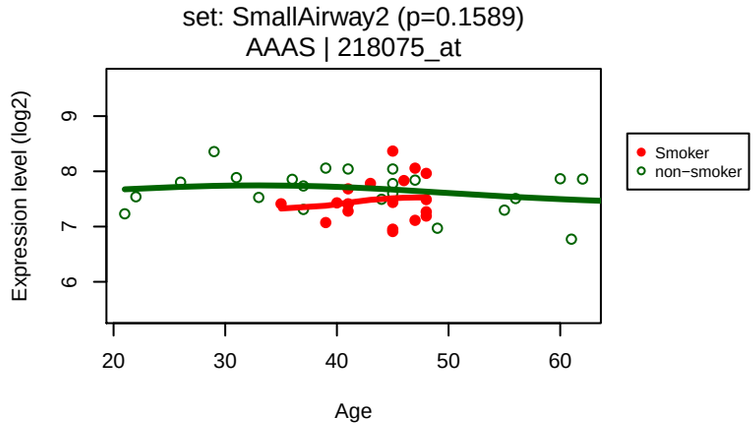
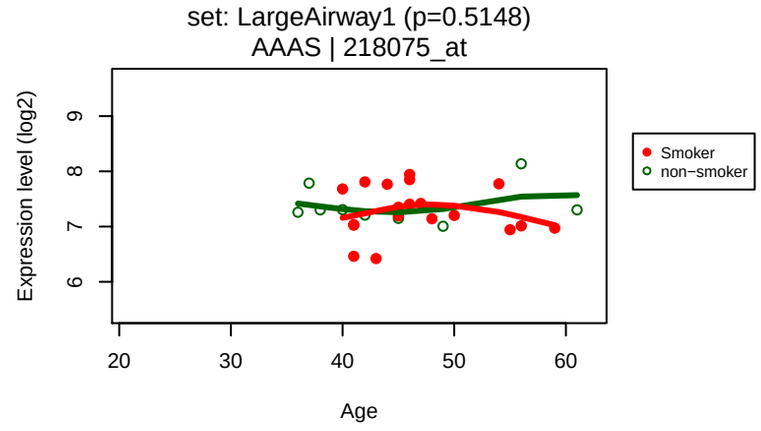

Gene expression of smokers and nonsmokers
gene: AAAS | 218075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.716
p=0.1589
p=0.4996
p=0.5148
p=0.8246
p=0.0134
p=0.325
p=0.0393
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

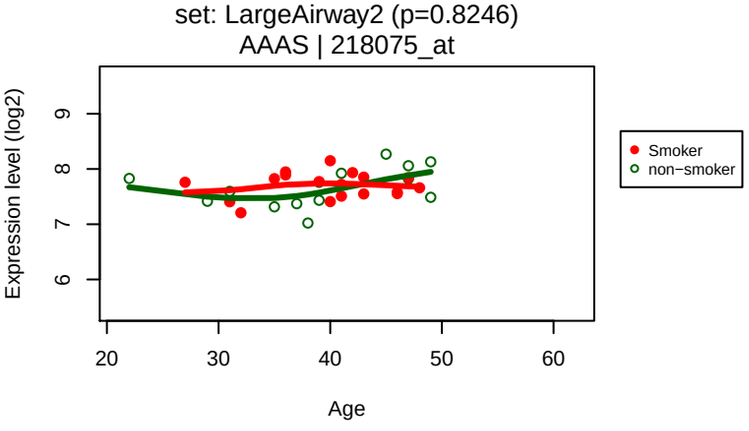

Gene expression ratio (smokers/nonsmokers)
gene: AAAS | 218075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.716
p=0.1589
p=0.4996
p=0.5148
p=0.8246
p=0.0134
p=0.325
p=0.0393
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

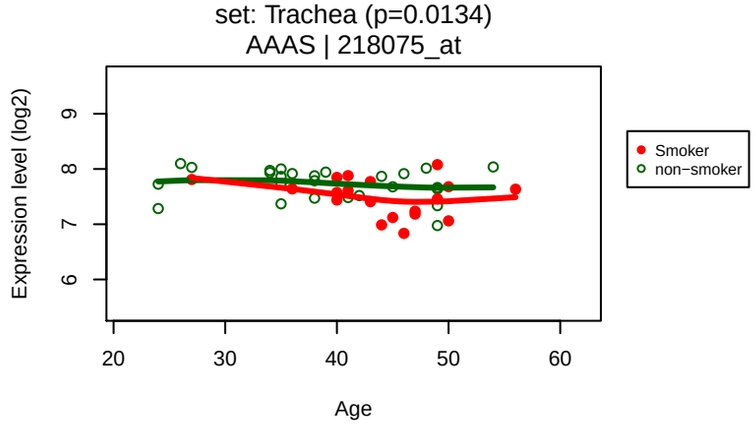

Gene expression ratio in Males
gene: AAAS | 218075_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0515
p=0.2111
p=0.0363
p=0.0207
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AAAS | 218075_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2898
p=0.1641
p=0.4504
p=0.9639
SmallAirway2
LargeAirway2
Trachea
Macrophage2

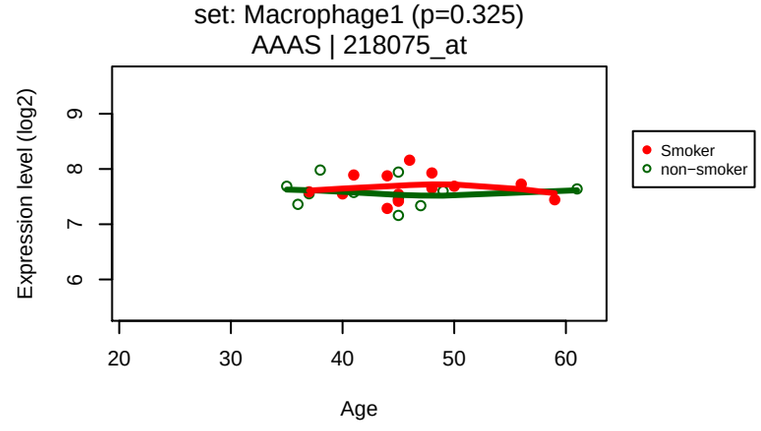

set: SmallAirway1 (p=0.716)
AAAS | 218075_at
Expression level (log2)
Age
Smoker
non−smoker

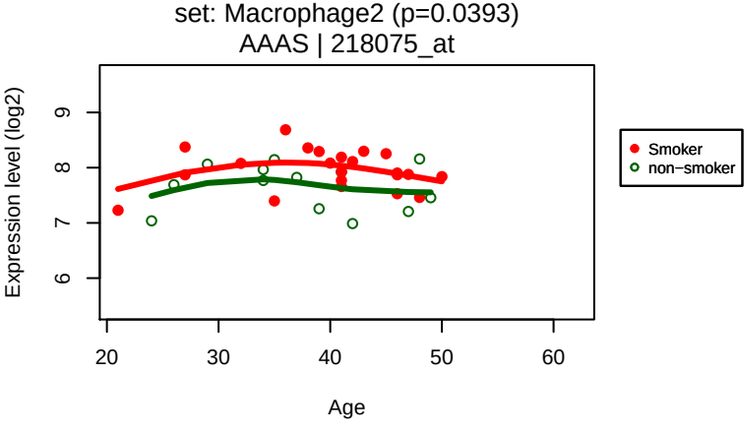

set: SmallAirway2 (p=0.1589)
AAAS | 218075_at
Expression level (log2)
Age
Smoker
non−smoker

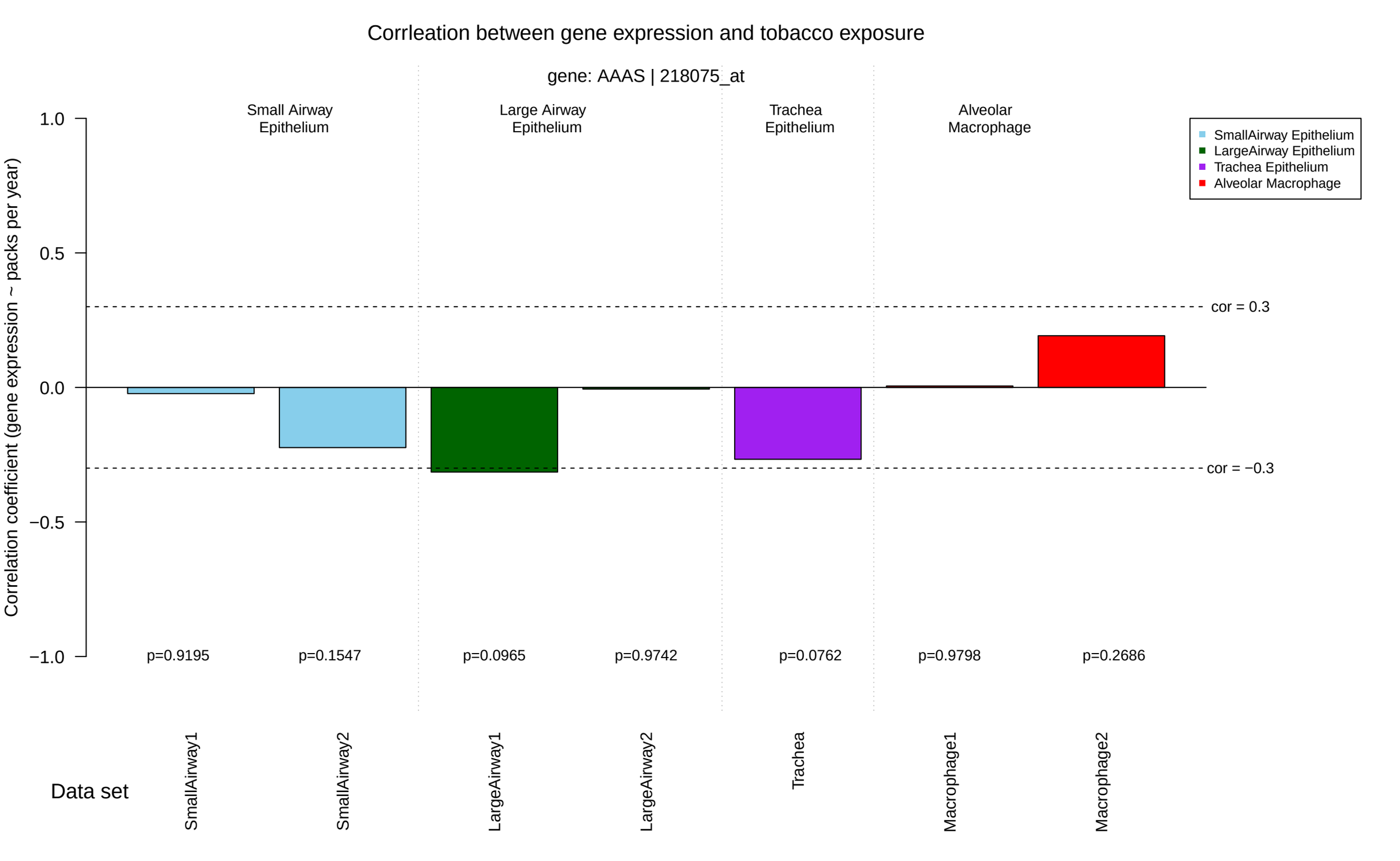

set: LargeAirway1 (p=0.5148)
AAAS | 218075_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8246)
AAAS | 218075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0134)
AAAS | 218075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.325)
AAAS | 218075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0393)
AAAS | 218075_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AAAS | 218075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9195
p=0.1547
p=0.0965
p=0.9742
p=0.0762
p=0.9798
p=0.2686
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2